

Ihar Petushkou

Linux Engineer | Python & AI Automation | Ansible, Terraform, AWS

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SUMMARY

Linux Engineer with 15+ years in software engineering, the recent years focused on Linux system administration, backend and automation. I set up, secure, troubleshoot and automate Linux servers (RHEL, Rocky Linux, Ubuntu, Debian), write Python and Bash that remove manual work, keep infrastructure as code with Ansible and Terraform on Amazon Web Services (AWS), and build AI agents on Claude Code and the Model Context Protocol (MCP) with tool allowlists, turn caps and cost limits.

SKILLS

Linux administration: RHEL, Rocky Linux, Ubuntu, Debian, Fedora, Alpine; server setup, hardening and migration; users, patching, cron, logs; systemd services and timers, boot and emergency-mode recovery; LVM, filesystems, swap; SELinux, AppArmor, firewalld, ufw, fail2ban, SSH hardening, TLS; Nginx and Caddy reverse proxies; KVM/QEMU, VPS, dedicated and on-prem servers

Python and backend: Python (FastAPI, Typer, Textual, asyncio, pytest, boto3), Bash; REST APIs, WebSockets, server-sent events; PostgreSQL, MySQL, Redis, MongoDB

Infrastructure and cloud: Ansible (roles, Vault), Terraform (modules, remote state), AWS (EC2, VPC, IAM, S3, RDS, Lambda, ECS, EKS, SQS, CloudWatch, Route 53, Secrets Manager), Docker, Docker Compose, Kubernetes, Helm, ArgoCD

CI/CD, monitoring and backups: GitHub Actions, GitLab CI, Jenkins; Prometheus, Grafana, Loki, Alertmanager; restore-tested backups

AI engineering: Claude Code for teams (CLAUDE.md, permission rules, hooks, subagents, skills, plugins), headless Claude Code in CI, Claude API and Claude Agent SDK, MCP servers (Streamable HTTP, OAuth), AI agents on OpenAI and Gemini, Ollama (private LLMs), RAG with pgvector

EXPERIENCE

Linux Engineer, Python & AI Automation | Freelance (Upwork), Remote | Aug 2025 – Present

- Provision, harden and troubleshoot Linux servers; automate them with Ansible and Python, and AWS environments with Terraform.
- Build AI agents on Claude Code that run engineering chores in CI (GitHub Actions) with tool allowlists, turn caps and cost limits, and MCP servers that give Claude a product's API and data.
- Deploy private AI on Ollama for data that must stay on the client's own servers.

Backend & DevOps Engineer, Mobile Development | Freelance (Upwork), Remote | Oct 2024 – Aug 2025

- Owned products end to end: Python and Dart backends, infrastructure as code (Terraform, Ansible) on AWS and Google Cloud, Docker and Kubernetes, automated release pipelines (CI/CD), LLM features in production.

Full-Stack Mobile Developer | Freelance (Upwork), Remote | Dec 2018 – Oct 2024

- Shipped cross-platform Flutter apps with the backends, Linux servers, databases (PostgreSQL, Firebase, Supabase) and App Store / Google Play release automation behind them, as sole owner.

JavaScript Developer | ISsoft, Minsk | May 2017 – Sep 2018

Web applications in JavaScript, React/Redux and TypeScript/Angular, with PHP, MySQL and PostgreSQL.

JavaScript / Python Developer, Team Lead | Itransition, Minsk | Jun 2014 – May 2017

Led production web applications and data visualization: React, D3, Node.js, Python/Django, MongoDB, on Linux.

Java Developer | IBA Group, Minsk | Jul 2010 – Oct 2013

Enterprise web applications in Java, GWT and Oracle DB: design, architecture, implementation, support.

PHP / JavaScript Developer | Avantime, Minsk | Jul 2005 – Dec 2007

Application design and implementation, MySQL optimization; Linux as a daily driver ever since.

PROJECTS

Norboten, an open-source platform for learning Linux by fixing broken machines | norboten.org, github.com/northelks/norboten

- Python terminal app that boots a real VM (QEMU/KVM, Lima), breaks it on purpose and grades the fix by checking the machine, rebooting it and checking again; 51 labs in 10 tracks (RHCSA, Linux, Bash, Python, Ansible, Docker, Terraform, AI agents, Claude Code, MCP), each proven solvable in CI.
- Server as code: FastAPI, PostgreSQL, Redis, Docker Compose, Ansible, Prometheus and Grafana, GitHub Actions with deploy and rollback; an MCP server with OAuth; 1,200+ automated tests.
- Claude Code agents run the chores in GitHub Actions, each with a turn cap, a tool allowlist and a measured cost; an AI tutor runs on the learner's own model and never sees the solution.

EDUCATION

Bachelor's degree, Computer Science and Software Technology, Belarusian State University of Informatics and Radioelectronics

CERTIFICATIONS AND LANGUAGES

Oracle Certified Professional, Java Programmer | English (B2, upper intermediate), Polish (A2), Belarusian and Russian (native)